

The Fracturing Stack: Global AI Fragmentation and What It Costs Us

*A Policy Whitepaper on Regulatory Divergence, Accountability Gaps,
and the Case for Managed Interoperability*

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Abstract

Global AI governance has fractured along jurisdictional, technical, epistemic, and accountability lines, producing a regulatory landscape that is neither safe nor coherent. This paper argues that fragmentation is not a transitional inconvenience but a structurally self-reinforcing dynamic: regulatory divergence creates incentives for jurisdictional arbitrage, arbitrage undermines audit quality, incompatible audits prevent mutual recognition, and the impossibility of mutual recognition forces duplicative compliance burdens that reward scale and concentrate market power, further reducing the diversity of AI systems under meaningful governance. The consequences are severe: cross-border AI deployments face compliance costs that EU and UK small and medium-sized enterprises estimate at €94,000–€453,000 annually per firm; accountability vacuums persist in exactly the contexts where AI systems make consequential decisions; and fragmentation makes collective learning from AI incidents structurally impossible. The dominant policy response — aspiring toward harmonization — is analyzed and rejected as both politically unrealistic and, in its lowest-common-denominator form, technically dangerous. This paper proposes instead a framework of managed interoperability: the development of compatible audit trails, mutual recognition of specific conformity assessments, shared incident reporting infrastructure, and federated liability allocation mechanisms. Drawing on analogies from aviation safety (ICAO), pharmaceutical mutual recognition agreements, and financial services equivalence regimes, the paper offers seven directed recommendations to G7 governments, the EU, standards bodies, and industry consortia. The window for managed interoperability is narrowing, not because political will is absent, but because technical lock-in is accelerating. This paper makes the case that the problem is urgent now.

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1. The Scenario That Exposes the Problem

Consider a concrete scenario — not hypothetical, but an increasingly ordinary description of how AI systems are built and deployed in 2026. A company headquartered in San Francisco develops a large language model. Its training dataset is partly composed of text scraped from European websites, which means it was generated by EU residents and processed, at least in part, under the jurisdiction of the General Data Protection Regulation. The model is fine-tuned on additional proprietary data held in Singapore, where the company maintains a regional operations hub and benefits from the city-state's relatively permissive data governance sandbox. The resulting AI system is deployed as a decision-support tool for financial services — specifically, it assists in credit-risk assessments affecting customers in the United Kingdom, Australia, and Nigeria. It is periodically audited by a third-party assurance firm operating under standards derived from the ISO/IEC 42001 framework, with elements drawn from NIST AI RMF 1.0. The system's outputs are used by human loan officers, but the outputs have material influence on final decisions. In two of the three jurisdictions, those loan officers are not required to disclose that AI contributed to the decision. In one, they are.

Now ask the governance question directly: what framework actually governs this system? The EU AI Act's extraterritorial provisions in Article 2 apply to providers and deployers whose outputs are intended to be used in the Union — but the system is not serving EU customers, even though it was trained on EU-originated data. UK regulators, in the post-Brexit period, are operating under a sectorally fragmented framework that has not yet produced binding AI-specific obligations for financial services AI of this type. Singapore's Model AI Governance Framework is voluntary. Nigeria has no dedicated AI legislation in force. The NIST AI RMF is explicitly voluntary and non-binding by design. The ISO/IEC 42001 audit provides a certificate of conformity with a management system standard, not a legal compliance determination.

So whose rules govern the failure when the system produces systematically biased credit assessments? Who bears liability when a Nigerian customer is wrongly denied a loan on the basis of a model trained partly in California on European data, fine-tuned in Singapore, and audited against American voluntary standards? The honest answer is: nobody knows. No jurisdiction has unambiguous authority. No liability framework maps cleanly to this value chain. No audit methodology was designed to be mutually recognized across these borders. And no shared incident database will capture what happened, analyze it, and feed the learning back into governance frameworks globally.

That is the problem this paper addresses. It is not an abstract policy concern. It is the structural condition under which a rapidly growing proportion of consequential AI deployments are currently operating.

Defining the Problem

Global AI fragmentation is not simply a policy inconvenience. It is a structurally self-reinforcing dynamic that creates compounding governance failures, raises the cost of compliance without raising its quality, and generates dangerous accountability vacuums precisely where AI systems are most consequentially deployed. The goal should not be harmonization — which is both unrealistic and potentially dangerous — but managed interoperability: a practical, achievable target state built on compatible audit infrastructure, mutual recognition of specific assessments, and shared incident reporting.

2. Defining Fragmentation Precisely

One reason the policy conversation about AI governance divergence moves in circles is that "fragmentation" is used as a catch-all term for at least four distinct but interacting phenomena. Most discussions treat these as separate problems that happen to coexist. This paper argues they are a single self-reinforcing system. Naming them precisely is the precondition for understanding how they interact.

2.1 Regulatory Fragmentation

Regulatory fragmentation is the most visible layer: different jurisdictions have enacted, or are enacting, different legal requirements, prohibited practices, definitional thresholds, and enforcement mechanisms for AI systems. The EU AI Act — Regulation (EU) 2024/1689 — establishes a mandatory risk-based classification system with binding requirements for high-risk applications and outright prohibition of certain practices. The United States has no equivalent federal statute as of May 2026, relying instead on a patchwork of executive orders, sector-specific agency guidance, and voluntary frameworks. China has enacted a suite of algorithm-specific and generative AI-specific measures under the authority of the Cyberspace Administration of China. Singapore, Japan, Canada, the United Kingdom, India, and Brazil are each at different stages of their own frameworks, often with different foundational concepts of what "high risk" means, who the responsible party is, and what transparency requires.

Regulatory fragmentation is not simply a matter of different rules: it is a matter of different rule-making logics. The EU's logic is *ex ante* — requirements must be met before a system is deployed. The US logic (where it exists at all) tends to be *ex post* — harms are addressed through enforcement and litigation after they occur. China's logic is a hybrid of prior approval requirements and ongoing content governance obligations. These are not minor implementation differences; they reflect fundamentally different theories of how technological risk should be governed. A company subject to all three must design compliance programs for three different epistemic models of regulatory intervention, not merely three different rule books.

2.2 Technical Fragmentation

Technical fragmentation operates at the level of standards, audit methodologies, data format requirements, and interoperability specifications. Multiple standards bodies — ISO/IEC JTC 1/SC 42, IEEE, NIST, ITU, ETSI, CEN/CENELEC — are developing AI-related standards in parallel, with overlapping scope and occasional direct conflicts. ISO/IEC 42001 specifies requirements for AI management systems; NIST AI RMF 1.0 provides a voluntary risk management framework with different structural logic; IEEE's P7000 series addresses ethics in system design; ETSI is developing standards tailored to the EU AI Act's conformity assessment requirements. Each of these bodies has its own governance structure, membership, process timeline, and relationship to regulatory mandates.

Data localization requirements compound the technical layer. China's Data Security Law and Personal Information Protection Law require that data generated within China be stored domestically and in many cases prohibit its transfer without a government security assessment. The EU's GDPR and successor frameworks restrict cross-border data flows to jurisdictions with adequate protection determinations or appropriate safeguards. The US has no equivalent federal data localization mandate but various sector-specific requirements (HIPAA, ITAR, FedRAMP) create de facto localization obligations for specific data types. A global AI training pipeline that would benefit from pooling data across these jurisdictions must navigate requirements that are not merely different but often directly incompatible — you cannot simultaneously store data in China, transfer it freely out of the EU, and subject it to US federal cloud security requirements.

2.3 Epistemic Fragmentation

Epistemic fragmentation is the deepest and least-discussed layer: different jurisdictions, and different regulatory traditions within jurisdictions, hold fundamentally different assumptions about what AI risk even is and who bears it. The EU framework is grounded in fundamental rights — risks are defined primarily by their potential to infringe dignity, privacy, and non-discrimination. The US framework, where it coheres at all, tends toward a consumer protection and national security frame — risks are defined primarily by market failures and strategic vulnerabilities. China's framework centers on social stability and national sovereignty — risks are defined primarily by potential for political disruption and information control. India's evolving governance guidelines articulate a "people first" and "innovation restraint" philosophy that is distinct from all three.

These are not merely rhetorical differences. They produce divergent judgments about what kinds of AI deployments require the most intense oversight, who counts as an affected party with standing to complain, what kinds of evidence should trigger regulatory action, and what a "safe" AI system means. An AI system certified as compliant in one epistemic frame may be categorically prohibited in another, not because the underlying technical facts differ, but because the normative premises that transform facts into risk assessments are incommensurable. This makes harmonization genuinely difficult in a way that regulatory convergence in, say, pharmaceutical safety is not — you can debate clinical trial methodology across cultures, but you cannot easily debate what counts as a legitimate use of facial recognition without

confronting foundational disagreements about civil liberties, collective security, and the role of the state.

2.4 Accountability Fragmentation

Accountability fragmentation is what makes all three other layers dangerous in practice. Even where liability frameworks exist within jurisdictions, they do not map cleanly to cross-border AI deployments. The EU AI Act creates obligations for providers and deployers, and extends its reach extraterritorially when outputs are intended for EU users. But it does not specify how liability is to be allocated when a provider is US-based, a deployer is Singapore-based, a sub-processor is UK-based, and the affected users are Nigerian. The emerging product liability frameworks in the EU create upstream obligations, but their practical reach across value chains that span multiple jurisdictions remains untested. In common law jurisdictions, tort liability for AI harm typically requires establishing causation between a specific system's output and a specific harm — increasingly difficult when AI systems are embedded in multi-stage pipelines where multiple vendors, fine-tuners, and deployers have each shaped the output.

The result is a liability topology riddled with gaps: value chain segments where no jurisdiction has clear authority, AI applications where no entity can be identified as the "provider" in a legally actionable sense, and incidents where the harm is clear but the causal attribution required to trigger any remedy is impossible to establish. These are not edge cases. They are the ordinary condition of frontier AI deployment in 2026.

2.5 A Single Self-Reinforcing System

The crucial point — and the one that most policy commentary misses — is that these four layers are not parallel problems. They are interconnected components of a single reinforcing system. Regulatory fragmentation at the legal layer creates ambiguity about which technical standards apply, which produces a proliferation of competing standards. Competing standards produce audit methodologies that are mutually unrecognized, which means compliance in one jurisdiction does not reduce compliance burden in another. The impossibility of mutual recognition forces duplicative compliance, which raises the cost of cross-border deployment, which creates incentives for regulatory arbitrage. Regulatory arbitrage undermines the epistemic integrity of audits. Untrustworthy audits make it politically impossible to negotiate liability allocation frameworks across borders, because no government will extend mutual recognition to a process it cannot trust. And the absence of shared liability frameworks makes accountability fragmentation structurally permanent. Each layer makes every other layer worse. This is why incremental reforms at any single layer have so consistently failed to produce systemic improvement.

3. The Major Fault Lines

3.1 The European Union AI Act

The EU AI Act — formally Regulation (EU) 2024/1689 of the European Parliament and of the Council, published in June 2024 — is the most comprehensive binding AI governance framework currently in force among major economies. It establishes a four-tier risk classification system: prohibited AI practices (Article 5), high-risk AI systems (Chapter III), limited-risk systems with transparency obligations (Article 50), and minimal-risk systems subject only to general principles. The regulation's high-risk category is defined by two routes: AI systems used as safety components in products subject to EU harmonization legislation under Annex I, and a specific list of application domains in Annex III — covering biometric identification, critical infrastructure, education, employment, essential services, law enforcement, migration management, and administration of justice.

For high-risk systems, the obligations are substantial. Providers must establish quality management systems, maintain technical documentation, implement data governance measures, ensure transparency and provide instructions for use, design for human oversight, achieve accuracy and robustness standards, and — where required by Annex I legislation — undergo third-party conformity assessment by a notified body before market placement. This last requirement is where a significant implementation challenge emerges: as of mid-2026, the EU's notified body infrastructure for AI conformity assessment is materially underdeveloped. Notified bodies must themselves be accredited by national accreditation authorities under Regulation (EC) No 765/2008, and the number of bodies with sufficient AI expertise to conduct meaningful assessments is far smaller than the volume of high-risk systems that will require assessment. The practical consequence is a bottleneck that will distort the market long before it is resolved.

The Act's extraterritorial scope, established in Article 2, extends its requirements to providers established in third countries to the extent that the output of their AI systems is intended to be used in the EU. This is a significant reach: a company with no EU presence that sells AI services to EU businesses is subject to the Act if those services influence decisions affecting EU persons. The "Brussels Effect" — the dynamic by which EU regulatory standards propagate globally because large companies find it efficient to design to the highest applicable standard — is real, but it has limits. The Act's third-country recognition mechanism, which would allow non-EU conformity assessment infrastructure to be recognized as equivalent, remains embryonic. Without functioning mutual recognition agreements, third-country providers must either route their conformity assessment through EU notified bodies (costly and slow) or risk operating in a compliance gray zone.

3.2 The United States: Sector-Specific and Voluntary

The United States has not enacted federal AI legislation as of May 2026. The Biden administration's Executive Order 14110 of October 2023, "Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence," directed agencies to develop sector-specific guidance and risk assessments, mandated safety testing reporting from developers of powerful dual-use foundation models, and tasked NIST with developing guidelines and best practices. It was a significant executive action, but it was not legislation, and the Trump administration's subsequent executive action in January 2025 explicitly reversed several of its provisions,

directing agencies to develop AI policy centered on US leadership and competitiveness rather than safety-first frameworks.

What remains is a patchwork. The FDA regulates AI as a medical device under existing statutory authority and has issued specific guidance on AI/ML-based software as a medical device (SaMD), including concepts for predetermined change control plans. The FTC exercises authority over deceptive or unfair trade practices involving AI, with increasing enforcement interest in algorithmic discrimination and AI-generated deception. The EEOC has issued guidance on AI in employment. The CFPB has addressed AI in credit. NIST's AI RMF 1.0, published January 2023, provides a voluntary, non-prescriptive framework organized around four core functions — Govern, Map, Measure, and Manage — that organizations can adopt at their own discretion and to whatever degree they find appropriate. State-level legislation, particularly in California, Texas, and Colorado, adds a further layer of inconsistency. For a company attempting to comply simultaneously with EU AI Act obligations and the US environment, the challenge is not navigating two coherent frameworks — it is navigating one binding, prescriptive, comprehensive framework and one fragmented, voluntary, sector-specific environment in which the operative requirements depend on which agency has jurisdiction over the relevant application domain and which state the deployer is incorporated in.

3.3 China: A Coherent Alternative Architecture

China's AI governance stack is internally coherent in a way that the US patchwork is not, but its coherence creates a different and arguably more severe interoperability problem: it is coherent within a value system that is incommensurable with the rights-based frameworks of the EU and the liberal democratic assumptions underlying most Western governance theory.

The architecture consists of several interlocking instruments. The Provisions on the Management of Algorithmic Recommendations (effective March 2022) govern the use of algorithmic recommendation systems — defined broadly enough to encompass most consumer-facing AI — requiring that providers register and file algorithmic information with the Cyberspace Administration of China (CAC), protect user rights, and ensure that recommendations do not create "addictive" behaviors or facilitate "bad information." The Provisions on the Administration of Deep Synthesis Internet Information Services (effective January 2023) govern synthetic media generation. Most significantly for present purposes, the Interim Measures for the Administration of Generative AI Services, promulgated jointly by the CAC and six other ministries and effective August 15, 2023, establish requirements for providers of public-facing generative AI services in China: prior security assessment and algorithm filing, training data lawfulness, content governance obligations requiring adherence to "socialist core values," and ongoing monitoring and complaint resolution systems.

Underlying this regulatory layer is a data governance stack — the Cybersecurity Law, the Data Security Law, and the Personal Information Protection Law — that creates comprehensive data localization and cross-border transfer restriction requirements. Data classified as "important data" or "core data" under China's data classification regime cannot leave the country without government approval. For AI developers, this means that training datasets assembled in China cannot be used to train models deployed globally, and conversely that global training runs

cannot incorporate China-originated data without navigating a security assessment regime designed with national sovereignty as its primary objective. There is no pathway to mutual recognition between this architecture and the EU or US frameworks, not because the technical problems are unsolvable, but because the underlying governance values are irreconcilable on the specific points that matter most to each framework.

3.4 Emerging Market Divergence

The governance challenge is not confined to the three major blocs. A growing set of jurisdictions — home to the majority of the world's population and increasingly significant AI deployment contexts — are developing their own frameworks in ways that create additional fault lines.

India's Ministry of Electronics and Information Technology (MeitY) released the India AI Governance Guidelines under the IndiaAI Mission in 2025, structured around seven foundational "sutras" — Trust, People First, Innovation Restraint, Fairness and Equity, Accountability, Understandable by Design, and Safety and Sustainability. The framework is explicitly voluntary at present and organized to encourage innovation alongside risk mitigation, drawing on India's Digital Public Infrastructure experience to propose governance models appropriate to a large, diverse, low-income-distribution context. India's framework is thoughtful and contextually appropriate, but it is neither aligned with EU requirements nor compatible with US sector-specific approaches. The African Union issued an AI Continental Strategy in 2024, creating a continental framework that emphasizes sovereignty, capacity building, and development-oriented AI applications. Brazil's draft AI Bill — modeled in significant respects on the EU AI Act but with modifications reflecting Brazilian constitutional traditions and development context — has been under parliamentary consideration and is expected to adopt a risk-based classification approach. Singapore's Model AI Governance Framework and AI Verify initiative represent a governance-by-testing approach that is innovative but explicitly positioned as complementary to, rather than harmonized with, major binding frameworks.

The risk in this proliferation is not merely that many frameworks exist. It is that smaller jurisdictions, facing resource constraints on governance capacity, face pressure to adopt one of the three major models wholesale — EU, US, or Chinese — without adequate assessment of whether the fit is appropriate. Adopting the EU AI Act's requirements without the EU's enforcement infrastructure and legal context creates compliance theatre. Adopting the US voluntary framework without the US's litigation environment and agency enforcement capacity creates no effective governance at all. And no jurisdiction outside China has seriously considered adopting China's model, given its political content requirements. The result is that emerging market AI governance tends to be either aspirational without implementation capacity, or borrowed without appropriate adaptation.

3.5 Standards Bodies: Converging in Rhetoric, Fragmenting in Practice

International standards bodies offer a promising but so far inadequate path toward technical convergence. ISO/IEC JTC 1/SC 42 — the joint technical committee on Artificial Intelligence under ISO and IEC — has published 42 standards and has 49 under development as of 2025, covering AI terminology, bias mitigation, explainability, robustness, data quality, risk

management, and conformity assessment. IEEE's P7000 series addresses ethics in system design. ITU's Focus Group on AI for Autonomous and Assisted Driving (FG-AI4AD) and related groups address sector-specific AI applications. NIST's AI RMF, though a national framework, has been widely referenced internationally. ETSI is developing standards specifically tailored to EU AI Act conformity assessment. CEN/CENELEC is working on harmonized European standards that will create a presumption of conformity with the Act for systems that comply with them.

In December 2025, ISO, IEC, and ITU issued a joint "Seoul Statement" committing to cooperation on AI standards, pledging to advance "AI standards that reflect global needs, support regulatory alignment, and foster interoperability, trust and inclusion." This was a meaningful political signal. It was not a technical solution. The three organizations retain different governance structures, different development processes, different membership compositions, and different relationships to their respective regulatory environments. The Seoul Statement represents a shared aspiration; it does not resolve the concrete conflicts between, for example, ETSI's EU-Act-specific conformity assessment standards and ISO/IEC 42001's global management system standard, which have overlapping but non-identical scope and requirements. Companies seeking compliance across multiple jurisdictions must currently map against all of these frameworks independently, because no authoritative mapping exists that a regulator in any jurisdiction will accept as a substitute for jurisdiction-specific compliance demonstration.

⚠ Critical Gap: Standards Convergence Without Regulatory Recognition

Even where standards bodies achieve technical alignment, convergence in standards text does not produce regulatory convergence if the associated conformity assessment frameworks — the bodies, the procedures, the acceptance criteria — remain jurisdiction-specific. A company that achieves ISO/IEC 42001 certification is not thereby compliant with the EU AI Act, the UK's evolving framework, or any other binding regulation. Standards convergence without mutual recognition infrastructure is insufficient. The Seoul Statement addresses the former; nobody has yet seriously addressed the latter.

4. How Fragmentation Compounds: The Reinforcing Loop

The most important analytical contribution this paper makes is to show not simply that fragmentation exists at multiple levels but how each level makes the others worse. Understanding this dynamic is essential to understanding why piecemeal reform keeps failing.

Begin with regulatory fragmentation. When jurisdictions enact different legal requirements for AI systems, they create a compliance landscape in which the minimum cost of cross-border deployment is the sum of compliance with each applicable regime. In principle, this creates an incentive to seek regulatory harmonization. In practice, however, it creates a different and more immediately attractive incentive: jurisdictional arbitrage. Companies with the resources to do so locate components of their AI value chains in jurisdictions with lower regulatory requirements for

those specific components. Model training occurs where data localization requirements are least restrictive. Deployment entities are structured in jurisdictions where liability exposure is minimized. Audit activities are performed against whichever standard is least demanding while still satisfying the most important regulatory market.

This arbitrage behavior is rational at the firm level and corrosive at the system level. It is rational because firms face real compliance costs and competitive pressure. An EU-funded study found that compliance costs constitute approximately 17.22 percent of the total cost to build an AI system subject to regulation — a figure that scales with the number of independent regimes to which a system must conform. The ACT | App Association survey found that EU and UK tech SMEs lose between €94,000 and €453,000 annually per firm from AI regulatory delays and compliance burdens. These are not trivial numbers for small organizations. Arbitrage reduces them. But the systemic effect is to hollow out the governance frameworks that regulatory fragmentation produced: when the most sophisticated actors can route around requirements, the populations actually subject to substantive oversight are those too small to arbitrage — startups, non-commercial deployers, public sector actors — while the largest commercial AI deployments, which pose the greatest systemic risk, operate in structured ambiguity.

Arbitrage, in turn, undermines audit quality. When organizations select their audit methodology based on which standard is least demanding rather than which provides the most meaningful accountability signal, the audit process becomes a compliance theater exercise rather than a genuine risk assessment. Third-party auditors competing for business have incentives to maintain relationships with the entities they audit. In the absence of a shared understanding of what a credible AI audit looks like — which is absent because of technical fragmentation at the standards level — audit quality is essentially unverifiable. A certificate of conformity with NIST AI RMF guidance, ISO/IEC 42001, EU AI Act requirements, or any combination of these tells a regulator in another jurisdiction essentially nothing about whether the audited system is actually safe or trustworthy by their standards, because the underlying audit methodology was not designed with their framework in mind.

The unverifiability of audits makes mutual recognition politically impossible. No government will agree to accept another jurisdiction's conformity assessment for AI systems deployed in consequential domestic contexts if it has no way to validate that the assessment was conducted to an appropriate standard. Without mutual recognition, every jurisdiction requires its own compliance demonstration. This means that companies deploying AI across multiple jurisdictions must conduct multiple parallel conformity assessments, maintain multiple documentation streams, interact with multiple regulators, and resolve multiple sets of potentially conflicting requirements — all simultaneously. The cost of doing this properly is high enough that only large organizations can absorb it without it becoming prohibitive. Smaller companies rationally exit markets. This is not a theoretical prediction: Apple withheld several AI Intelligence features from the EU market in 2024 and 2025 due to regulatory uncertainty; numerous smaller AI companies have explicitly excluded EU or UK markets from their initial deployment scope. The companies that remain in the most regulated markets tend to be the largest, because only they can afford the compliance overhead.

Market concentration resulting from compliance cost barriers reduces the diversity of AI systems subject to meaningful governance. Fewer, larger providers means that the AI systems making consequential decisions across a jurisdiction are developed by a smaller set of organizations with more concentrated value chain architectures and more similar technical approaches. Governance frameworks designed with an implicit assumption of market diversity — where multiple competing systems create mutual accountability — are poorly equipped for a market dominated by a small number of very large providers. And concentrated providers, operating at scale across jurisdictions, have both the incentive and the capability to participate in standards-setting processes in ways that shape technical standards to their existing architectures, which further entrenches the technical fragmentation that made concentration possible in the first place.

The loop closes: technical fragmentation → competing standards → unverifiable audits → regulatory arbitrage → market concentration → capture of standards processes → reinforced technical fragmentation. Each cycle of this loop bakes in incompatibilities that become harder to bridge, because the economic interests that benefit from the current structure grow stronger with each iteration.

Table 1: Comparative Summary of Major Jurisdictional AI Governance Frameworks

Dimension	European Union (AI Act)	United States (Patchwork)	China (Layered Stack)	United Kingdom (Post-Brexit)	India (AI Governanc e Guidelines)	Singapore (Model AI Gov.)
Primary Instrument	Regulation (EU) 2024/1689	EO 14110 (partly revoked); sector-specif ic agency rules; NIST AI RMF 1.0 (voluntary)	Algorithm Recommendations Provisions (2022); Generative AI Measures (2023); Data Security Law	Pro-innovati on framework; sector-specif ic regulator guidance (FCA, ICO, CMA)	India AI Governance Guidelines (MeitY, 2025); voluntary	Model AI Governance Framework (IMDA, voluntary); AI Verify toolkit
Binding / Voluntary	Binding	Mostly voluntary; sector-specif ic binding	Binding	Sector-speci fic; no general	Voluntary (as of 2025)	Voluntary

		rules (FDA, FTC, CFPB)		binding AI law yet		
Governance Logic	Ex ante risk classification ; fundamental rights-based	Ex post enforcement ; market failure and national security frame	Prior approval plus ongoing content governance; social stability and sovereignty frame	Pro-innovation; sector regulator discretion	Innovation-first; people-centric principles	Innovation-enabling; accountability through testing
Risk Classification	Prohibited / High-Risk / Limited-Risk / Minimal-Risk	No general classification ; sector-specific risk categories (e.g., FDA SaMD)	Public-opinion attributes / social mobilization capability triggers; generative AI separately regulated	No formal classification ; sector regulators define risk contextually	Risk mitigation recommendations by sector; classification under development	No formal classification ; governance by use-case and testing
Extraterritorial Reach	Strong: applies to providers/deployers whose outputs target EU users (Art. 2)	Limited: sector-specific jurisdiction; state laws vary	Strong within China: applies to any service accessible in PRC; data localization prevents outflow	Moderate: GDPR-equivalent reach for data; AI rules evolving	Limited	Minimal
Conformity Assessment	Mandatory third-party for highest-risk; self-assessment for	Voluntary (NIST RMF); mandatory FDA pre-market	Security assessment and algorithm filing with CAC required for	No mandated AI conformity assessment; sector regulator	Voluntary; AI Safety Institute to provide guidance	AI Verify toolkit; voluntary self-testing

	most Annex III systems	review for medical AI	public-facing services	approval varies		
Data Governance	GDPR applies to training data; adequacy decisions govern cross-border transfers	Sector-specific (HIPAA, COPPA, CCPA/CPR A); no federal comprehensive data law	Data Security Law; PIPL; cross-border transfer security assessments; localization mandated	UK GDPR; adequacy decisions post-Brexit	DPDP Act (2023); cross-border transfer framework under development	PDPA; moderate data localization
Maximum Penalty	Up to 7% global annual turnover or €35M (GPAI); 6% or €30M (high-risk)	FTC: civil penalties up to \$51,744/day per violation; sector variation	Warning, fines up to RMB 1M; service suspension; criminal referral	ICO: up to £17.5M or 4% global turnover (data); AI enforcement evolving	Not yet specified	Not applicable (voluntary)
Standards Alignment	CEN/CENELEC harmonized standards; ISO/IEC 42001 referenced; ETSI developing	NIST AI RMF primary; ISO/IEC JTC 1/SC 42 referenced but not mandated	GB/T national standards; some ISO alignment; separate national standardization	DSIT references ISO/IEC 42001 and NIST RMF; no mandated standards	ISO/IEC and NIST referenced; national standards under development	ISO/IEC 42001; own AI testing standards
Liability Framework	AI Liability Directive proposed (still under negotiation as of 2026); product	Common law tort; section 230 questions; no AI-specific liability statute	Provider primary liability; deployer secondary; user liability acknowledged	Common law; sector-specific; Law Commission review ongoing	Not yet established	Not established

	liability amendment					
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5. What Fragmentation Costs — Specifically

5.1 Compliance Costs

The direct financial costs of AI regulatory fragmentation are significant and growing, though they are unevenly distributed in ways that have structural market consequences. For large enterprises with dedicated regulatory affairs functions and global legal teams, the marginal cost of adding another jurisdiction's AI compliance requirements to an existing program is high but manageable. For small and medium-sized enterprises — which represent the majority of AI developers and deployers by number — the costs are often prohibitive. Survey data from ACT | The App Association, covering more than 1,000 technology MSMEs across the EU, UK, and US in 2024, found that EU and UK tech startups and SMEs lose on average between €94,000 and €453,000 annually per firm from delayed AI model launches and compliance burdens; for the most directly affected small-tech firms, the annual impact rises to between €160,000 and €528,000. An EU-funded study found that compliance costs constitute approximately 17.22 percent of the total cost to build an AI system subject to regulation.

These figures capture the within-jurisdiction burden for one of the world's most comprehensive frameworks. They do not capture the additional costs of multi-jurisdictional compliance. A company seeking to deploy an AI system simultaneously in the EU, UK, US, Singapore, and India must maintain compliance programs calibrated to five different governance frameworks, five different documentation requirements, five different liability structures, and — depending on the application — multiple sectoral regulators within each jurisdiction. The marginal cost of each additional jurisdiction is not zero even for large firms, and for small firms it is frequently the decision variable that determines whether they enter a market at all.

5.2 Innovation Chilling Effects

The deployment contexts that get avoided are precisely those that most need AI governance. High-stakes application domains — medical AI, judicial decision support, social welfare administration, financial access — attract the most regulatory scrutiny and generate the most complex compliance requirements. They are also the domains where AI could deliver the greatest social benefit and where the consequences of ungoverned deployment are most severe. The innovation chilling effect of fragmentation is therefore asymmetric: it depresses deployment in high-scrutiny contexts (where governance would be most valuable) while leaving low-scrutiny contexts (where the risks are often lower) relatively unaffected. The ACT | App Association survey found that approximately 60 percent of EU and UK technology startups face delayed access to frontier AI models and features, and more than one-third are forced to strip or downgrade functionality to comply with regulatory requirements.

This creates a perverse dynamic: the regulatory frameworks whose purpose is to make high-risk AI deployment safer are, in practice, making high-risk AI deployment less accessible to developers who would be most responsive to governance requirements, while large incumbents who can absorb compliance costs enter these markets on terms that shape them to their existing product architectures. The regulatory burden intended to protect people in high-stakes contexts instead serves primarily as a barrier to entry that entrenches incumbent advantage — without necessarily making the AI systems that do enter those contexts meaningfully safer.

5.3 The Accountability Vacuum

Fragmentation creates accountability vacuums in exactly the deployment contexts where accountability matters most: cross-border AI applications affecting consequential decisions. The scenario described in the opening section — a credit AI trained in one jurisdiction, deployed in another, affecting users in a third — illustrates a structural pattern. No jurisdiction has unambiguous authority over the full value chain. Liability frameworks designed for within-jurisdiction deployments do not map to distributed AI value chains. The entity with the most influence over the system's risk characteristics — the foundation model developer — is typically in a different jurisdiction from the entity that deployed the system, which is typically in a different jurisdiction from the persons affected by it.

This is not a hypothetical problem awaiting future AI deployments. As of 2026, AI systems are routinely used in consequential decision contexts — credit assessment, insurance underwriting, predictive policing, visa adjudication, medical diagnosis support, benefits eligibility — with cross-border value chains that render jurisdiction unclear. McKinsey's State of AI survey found that nearly half of organizations have already reported at least one significant negative consequence from AI use, yet governance responses to those consequences are episodic and jurisdictionally constrained. NTIA's 2024 AI Accountability Policy Report identified information flow, independent evaluation, and consequence mechanisms as the three pillars of accountability — and noted that all three are systematically compromised in cross-border deployments.

5.4 The Security Dimension

Regulatory fragmentation creates exploitable security gaps in AI system oversight that are particularly dangerous for dual-use and critical infrastructure applications. When different jurisdictions apply different security standards to AI systems used in critical infrastructure, the weakest link in a globally interconnected network determines effective security for all of it. A power grid management AI system that meets rigorous security requirements in one country but is connected to components certified to different standards in an adjacent country creates attack surfaces that sophisticated adversaries will identify and exploit. The absence of mutual recognition frameworks means that security certifications obtained in one jurisdiction create no assurance in another — and no obligation to disclose security incidents across borders, which would be the minimum requirement for collective defense.

The dual-use dimension of AI fragmentation is perhaps the most underappreciated security concern. AI capabilities developed for civilian applications — computer vision, natural language

processing, sensor fusion, decision optimization — have obvious military and intelligence applications. In a fragmented regulatory environment, the same capability may be tightly controlled in one jurisdiction (subject to export controls, security assessments, and classified application restrictions) and freely available in another. This creates pathways for capability acquisition that circumvent the intent of export control regimes without technically violating their text, because the regulatory architecture for AI dual-use control is national rather than multilateral and poorly synchronized even among close allies.

5.5 The Epistemic Cost

Perhaps the least discussed but most consequential cost of fragmentation is epistemic: it makes collective learning from AI incidents structurally impossible. When an AI system causes harm, the ability to understand what went wrong, why it went wrong, and how to prevent similar failures elsewhere depends on sharing information about the incident across the relevant communities of developers, deployers, auditors, and regulators. In domains with mature incident reporting frameworks — commercial aviation, pharmaceutical adverse event reporting, nuclear safety — incident data is systematically collected, analyzed, and fed back into safety standards and operational practice in ways that improve safety over time even as the underlying technology evolves. AI has no equivalent infrastructure.

Regulatory fragmentation makes this infrastructure impossible to build. Incident reports filed with the EU AI Act's market surveillance authorities under Chapter IX will not automatically be shared with the UK's sector regulators, NIST, or the Singapore IMDA. There is no shared incident database, no common incident taxonomy, no cross-border reporting obligation, and no institution with the mandate and authority to aggregate and analyze AI incidents globally. The result is that every jurisdiction learns only from incidents that occur within its own regulatory visibility — which, given the cross-border nature of most significant AI deployments, is a severely truncated sample. We are individually ignorant in ways that make us collectively dangerous.

Table 2: Fragmentation Layer Analysis Matrix

Fragmentation Type	Mechanism	Key Manifestation	Effect on Other Layers	Current Gap	Minimum Viable Fix
Regulatory	Different legal requirements, prohibited practices, definitional thresholds	EU mandatory ex ante assessment vs. US voluntary ex post enforcement	Creates incentives for jurisdictional arbitrage; undermines standards convergence by producing	No binding multilateral AI governance instrument; no mutual recognition of	Bilateral regulatory cooperation agreements with information sharing; G7 common

	across jurisdictions	vs. China prior filing	jurisdiction-specific requirements	regulatory determinations	principles with teeth
Technical	Competing standards bodies, incompatible audit methodologies, data format conflicts, localization requirements	ISO/IEC 42001 vs. NIST RMF vs. ETSI EU-Act standards; China GB/T national standards	Prevents mutual recognition; forces parallel documentation; enables arbitrage toward weakest methodology	No authoritative cross-jurisdictional standards mapping accepted by regulators; notified body infrastructure underdeveloped	Regulator-endorsed mapping of ISO/IEC 42001 to jurisdiction-specific requirements; shared conformity assessment protocol
Epistemic	Different underlying conceptions of AI risk, who bears it, and what safety means	EU fundamental rights frame vs. US national security/market frame vs. China social stability frame	Makes harmonization politically impossible on core value questions; produces incommensurable risk assessments	No cross-jurisdictional forum for epistemic translation; risk definitions remain nationally siloed	OECD-facilitated working group on common risk taxonomy; not values harmonization but definitional interoperability
Accountability	Liability structures that do not map to cross-border AI value chains	EU AI Liability Directive still in negotiation; US tort law requires causation proof in multi-vendor AI chains; no cross-border liability allocation framework	Reinforces all other layers by removing consequences that would otherwise incentivize compliance; creates deployment without accountability	No federated liability allocation mechanism; no shared AI incident database; no cross-border enforcement cooperation framework for AI	Model bilateral treaty on AI liability allocation; mandatory cross-border incident disclosure for high-risk AI applications

Systemic (compounding)	Each layer's failures amplify the others, creating a reinforcing loop	Regulatory arbitrage → audit hollowing → mutual recognition impossible → compliance cost → market concentration → standards capture → reinforced fragmentation	Self-referential: the system as a whole resists reform at any single layer	No institution with mandate to address the system-level dynamic; all current governance efforts are layer-specific	Dedicated multilateral AI governance coordination body with cross-layer mandate; not a new treaty organization but a coordination function within an existing one
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6. Why Harmonization Is the Wrong Goal

There is a dominant policy instinct, expressed across think tank reports, regulatory consultations, and diplomatic communiqués, that the solution to AI governance fragmentation is harmonization — convergence on a single global framework, or at least a set of common principles binding enough to function like one. We want to address this directly, because we think it is wrong, and because pursuing it as a goal wastes political and institutional capital that could be directed toward something achievable.

Full harmonization would require one of two outcomes. Either one jurisdiction's governance values dominate the global framework — a scenario that is both politically unacceptable and epistemically dangerous — or the process of negotiation produces a lowest-common-denominator outcome that satisfies all parties precisely because it requires little of any of them. Neither outcome is desirable. The first would represent a form of governance imperialism: if the EU's fundamental rights frame is imposed globally, jurisdictions with different constitutional traditions and development contexts are required to adopt governance values they have not democratically chosen. If China's social stability and content governance frame is imposed globally, the result is a planetary-scale censorship and surveillance infrastructure. If the US's market-first, sector-specific frame is imposed globally, the result is the absence of meaningful governance for the most consequential AI applications.

The "Brussels Effect" — the well-documented tendency of EU regulations to propagate globally because large companies find it efficient to build to the highest applicable standard — is real and has produced genuine benefits in areas from data protection to chemical safety. But its limits in AI governance are already visible. The EU AI Act has already prompted market exits and feature withholding; it has created compliance burdens that small developers describe as prohibitive; and its extraterritorial reach has been met with explicit resistance from governments that view it as an imposition of EU values without EU democratic legitimacy. The Brussels Effect works well when the underlying values embedded in regulation are broadly shared and the main

barrier is coordination, not genuine disagreement. In AI governance, the genuine disagreements are fundamental: about privacy, autonomy, state power, and what kind of future we want AI to help build. Regulatory imperialism is not a solution to those disagreements; it simply relocates the conflict from a negotiating table to a compliance enforcement proceeding.

The lowest-common-denominator alternative is even worse. A global AI governance framework negotiated to the point of consensus among the EU, US, China, India, and other significant parties would be precisely as ambitious as all parties can simultaneously accept — which means it would be roughly as demanding as the least demanding party's position on each question. That is, it would be a framework with voluntary risk management guidance, no binding conformity assessment requirements, content governance obligations reflecting the broadest consensus on prohibited content (a very small set), and no effective enforcement mechanism. This would provide the political satisfaction of a signed multilateral instrument while delivering essentially no improvement in actual AI governance. The OECD AI Principles, adopted in 2019 and now endorsed by more than 46 countries, represent roughly this outcome: a valuable signal of shared aspiration that has had limited impact on the divergent national frameworks that continued to proliferate after its adoption.

The better frame is managed interoperability. The goal is not that all jurisdictions govern AI in the same way — it is that AI systems built and deployed across jurisdictions can be meaningfully governed at all, that accountability can be assigned, that incidents can be learned from, and that compliance requirements can be satisfied without the duplicative overhead that currently makes cross-border deployment so expensive. Managed interoperability is technically and institutionally achievable in a way that harmonization is not, because it does not require agreement on values — only agreement on specific, bounded interface standards between governance systems that can otherwise remain distinct.

► The Managed Interoperability Argument

Managed interoperability means that governance systems developed by sovereign jurisdictions can exchange accountability information, recognize specific conformity assessments, and allocate liability without converging on identical rules or underlying values. It is the difference between asking two rail networks to use the same trains (harmonization) and building a standard gauge junction that allows trains from both networks to cross the border (interoperability). The junction is technically demanding but politically feasible. The same train is neither.

7. What Managed Interoperability Requires

Managed interoperability is not a vague aspiration — it has concrete technical and institutional requirements. We can describe what a viable target state looks like by examining where analogous challenges have been addressed in other domains, identifying what exists in AI governance, and specifying what is missing.

The most instructive analogy is commercial aviation safety under the International Civil Aviation Organization (ICAO). Aviation involves extraordinarily complex, safety-critical systems built by

manufacturers in a small number of countries, operated by airlines in nearly every country, flying across national boundaries continuously. Yet a passenger boarding a flight in Lagos connecting through Singapore to London has a reasonable expectation that every aircraft they board, regardless of where it was manufactured or registered, meets safety standards that are mutually recognized and meaningfully enforced. This is not because all aviation regulators agree on every aspect of aircraft safety — they do not — but because ICAO's Standards and Recommended Practices (SARPs) define the interface standards that enable bilateral and multilateral airworthiness recognition agreements to function. The FAA and EASA recognize each other's aircraft type certificates through a specific bilateral aviation safety agreement that does not require the FAA and EASA to have identical regulations, only to maintain trust in each other's certification processes as applied to specific agreed categories of determinations.

Pharmaceutical mutual recognition agreements (MRAs) offer a second analogy. The EU, the US, Canada, Australia, Japan, and several other jurisdictions have negotiated MRAs under which inspections of pharmaceutical manufacturing facilities by one jurisdiction's authority are recognized by others for specified purposes. These MRAs do not require that all parties have identical pharmaceutical regulations — they require that the inspecting authority's process, competence, and quality system be assessed and found adequate by the recognizing authority. The MRA covers the inspection process, not the underlying regulatory requirements, which remain jurisdiction-specific. This is exactly the structure needed for AI: mutual recognition of conformity assessment processes, not mutual adoption of regulations.

Financial services equivalence regimes provide a third partial analogy. The EU grants equivalence determinations to third-country financial regulatory frameworks under various directives, allowing third-country firms to access EU markets on the basis of home-country regulation that is assessed as equivalent in effect. These determinations are politically fraught — as the UK's experience post-Brexit demonstrates — but they demonstrate the institutional architecture: a process by which jurisdiction A assesses whether jurisdiction B's framework produces equivalent outcomes on specific dimensions, without requiring B to adopt A's rules.

What would these analogies look like in AI governance? We can identify five specific infrastructure components that are currently absent and would constitute the minimum viable architecture for managed interoperability.

First, a shared AI incident reporting and database infrastructure. No cross-border AI incident database exists. The EU AI Act creates national market surveillance authority reporting obligations, but these feed into national databases, and there is no institutional mechanism for cross-border aggregation even among EU member states, let alone with third countries. Building shared incident infrastructure would require agreement on a common incident taxonomy — not identical definitions of harm, but a shared classification system that allows incidents reported under different national frameworks to be meaningfully compared and aggregated. The Aviation Safety Reporting System (ASRS) operated by NASA provides a useful model: confidential, non-punitive reporting that produces system-level learning without creating liability exposure for reporters.

Second, a cross-border conformity assessment recognition framework for AI. Modeled on pharmaceutical MRAs, this framework would establish a process by which regulators in one jurisdiction assess the competence and process quality of conformity assessment bodies in another jurisdiction for specific categories of AI system assessment. A company that has obtained a conformity assessment for a high-risk AI system from a body recognized under this framework in its home jurisdiction would not need to repeat the assessment in the recognizing jurisdiction. This would not eliminate jurisdiction-specific requirements — only the costly duplication of the assessment process itself. The technical precondition is the development of a shared auditor competence framework and a shared assessment methodology for the assessment process itself (meta-audit). Neither exists currently.

Third, a common AI audit trail standard. For mutual recognition to function, the documentation produced by AI conformity assessment processes in different jurisdictions must be sufficiently compatible that a regulator in one jurisdiction can read and evaluate it. This requires not identical documentation, but a common data model for the key elements of an AI system audit record: training data provenance, model performance metrics, bias assessments, human oversight mechanisms, incident history, and post-market monitoring plans. ISO/IEC 42001 provides some of this structure but is not recognized as legally adequate documentation in any mandatory regulatory framework. A cross-jurisdictional effort, potentially through ISO/IEC JTC 1/SC 42 with explicit regulatory endorsement from multiple jurisdictions, to develop a recognized common audit trail standard would address this gap.

Fourth, a federated liability allocation mechanism for cross-border AI harms. This is the hardest component and the one furthest from existence. A federated mechanism would establish, through bilateral or multilateral treaty, the rules for determining which jurisdiction's liability framework applies to AI harms involving multi-jurisdictional value chains, how liability is allocated among value chain participants across borders, and what enforcement cooperation mechanisms exist to give cross-border liability determinations effect. The starting point is a model bilateral treaty — a template instrument that any two willing jurisdictions can adopt — rather than a comprehensive multilateral framework, which would take a decade to negotiate. The model treaty approach has precedent in tax (the OECD Model Tax Convention) and investment (the UNCTAD model bilateral investment treaty). There is no equivalent for AI liability.

Fifth, a permanent multilateral coordination function on AI governance. Not a new treaty organization — there is no appetite for one — but a standing secretariat function, most plausibly located within the OECD, ITU, or a dedicated G7/G20 working group, with a cross-layer mandate: tracking regulatory divergence, identifying emerging conflicts, facilitating bilateral negotiations on specific interoperability components, and publishing annual convergence assessments. The Global Partnership on AI (GPAI), established in 2020, aspired to something like this function but has lacked institutional weight and cross-layer mandate. The AI Seoul Process, initiated at the AI Safety Summit at Bletchley Park in November 2023 and continued through subsequent ministerial meetings, has produced useful political signals but no institutional architecture with operational capacity.

► What Exists vs. What Is Needed

As of May 2026, the global AI governance landscape has: OECD AI Principles (2019, endorsed by 46+ countries; non-binding); GPAI (multilateral; useful research function; no regulatory authority); G7 Hiroshima AI Process (ministerial-level; produced code of conduct for generative AI; no binding effect); ISO/IEC JTC 1/SC 42 (active standards development; 42 published standards; not legally recognized in most jurisdictions); Seoul Statement (December 2025; joint ISO/IEC/ITU commitment; aspirational). What does not exist: shared AI incident database; cross-border conformity assessment recognition framework; common AI audit trail standard; federated liability allocation mechanism; standing operational coordination secretariat.

8. Recommendations

The following seven recommendations are directed at specific actors and describe specific actions. Each is accompanied by a minimum viable version — the lowest-threshold version of the action that would constitute genuine progress. We are not in the business of recommending ideal outcomes that will not be achieved in the relevant timeframe. We are in the business of identifying the next achievable step.

Recommendation 1: G7 governments should establish a binding AI incident notification protocol. The minimum viable version is a mutual obligation among G7 governments to notify each other within 72 hours of any significant AI incident affecting critical infrastructure or involving a system with cross-border deployment, using a common incident taxonomy developed by and endorsed by all seven governments. This does not require full incident data sharing or harmonized definitions of harm — it requires only that incident existence and basic classification be shared in a timely way. The G7 Hiroshima AI Process provides the existing diplomatic infrastructure; the Seoul ministerial process provides the political context. What is missing is a binding commitment with a specified timeline and taxonomy. The actor is the G7 presidency, which rotates; the 2026 Canadian presidency should be used to achieve this.

Recommendation 2: The EU and UK should jointly initiate AI conformity assessment mutual recognition negotiations, beginning with medical AI. Medical AI is the most mature regulatory domain for AI governance in both jurisdictions — the EU's medical device regulation framework and the UK's MHRA have well-developed approaches to software as a medical device, and both jurisdictions have notified/designated bodies with relevant expertise. A UK-EU AI conformity assessment MRA, modeled on the existing pharmaceutical MRA, would establish a precedent and institutional template for broader AI mutual recognition. The minimum viable version is agreement to recognize each other's conformity assessments for Class IIa and above AI medical devices. The actor is the EU Commission (DG SANTE and DG CONNECT) and the UK DSIT/MHRA.

Recommendation 3: ISO/IEC JTC 1/SC 42, with explicit co-endorsement from the EU, UK, and US regulatory bodies, should develop and publish a common AI audit trail data standard. The standard should specify the minimum data elements required in an AI conformity assessment record for the record to be recognizable and interpretable by regulators in any endorsing jurisdiction. It should be expressed as a data schema, not a narrative standard, to enable machine-readable exchange. The minimum viable version is agreement on the data

elements and schema for high-risk AI systems in three application domains — medical, employment, and financial services — as a pilot, with expansion following. The actor is ISO/IEC JTC 1/SC 42 with regulatory co-endorsement from the EU AI Office, UK DSIT, and NIST.

Recommendation 4: NIST should develop and publish a regulator-endorsed mapping of the AI RMF to EU AI Act high-risk system requirements, in cooperation with the EU AI Office. This mapping would not assert that RMF compliance equals EU AI Act compliance — it cannot, because the RMF is voluntary and the Act is binding — but it would specify which RMF practices, implemented at which profile levels, satisfy which EU AI Act obligations for which system categories. This would materially reduce the documentation overhead for US-based companies seeking EU compliance and for EU companies seeking to understand their US obligations. The minimum viable version is a mapping covering the top five Annex III high-risk categories. The actor is NIST in cooperation with the EU AI Office, which was formally established in February 2024 within the European Commission.

Recommendation 5: The OECD should develop a model bilateral treaty on AI liability allocation for cross-border deployments, for voluntary adoption by member states. Modeled on the OECD Model Tax Convention, this instrument would provide a negotiated template specifying default rules for jurisdiction, choice of law, liability allocation among value chain participants, and enforcement cooperation for AI harms involving cross-border value chains. It should be designed for bilateral adoption — two willing governments can sign it — rather than requiring multilateral consensus. The minimum viable version is publication of the model instrument with commentary, analogous to the OECD Commentary on the Model Tax Convention. The actor is the OECD's Science, Technology and Innovation directorate, with input from the OECD's legal affairs office.

Recommendation 6: Industry consortia, led by existing bodies such as the Partnership on AI and the ITU AI for Good platform, should establish a shared AI incident and near-miss database, modeled on the Aviation Safety Reporting System (ASRS). The database should be voluntary, confidential, non-punitive, and governed by a multi-stakeholder board. Submissions should follow a common taxonomy developed in coordination with Recommendation 1. The minimum viable version is a pilot program covering AI systems in healthcare and financial services in participating jurisdictions, with a commitment to expand based on pilot findings. Unlike a regulatory database, an industry-led confidential database does not require legislative action — it requires agreement among industry participants on governance and resources.

Recommendation 7: The G7, in coordination with the OECD, should establish a standing AI Governance Coordination Secretariat with an explicit cross-layer mandate. The Secretariat's function is not to produce new governance frameworks — it is to track regulatory divergence, identify emerging conflicts, facilitate bilateral negotiations on specific interoperability components, and publish annual convergence assessments against a defined managed interoperability benchmark. It should have a small permanent staff and rotating secondments from member government regulatory bodies, standards organizations, and civil society. The minimum viable version is a joint G7 resolution establishing the function, its mandate, and its

funding formula, with a secretariat housed within the OECD. The 2026 Canadian G7 presidency provides a near-term opportunity.

9. Conclusion: The Window Is Narrowing

We want to close with a direct statement about urgency, because the framing of AI governance debates often implies that these are long-horizon problems. They are not. The window for managed interoperability is closing, not because political will is absent — in fact, the political will for AI governance cooperation has never been higher — but because technical lock-in is accelerating in ways that make bridging increasingly expensive with each passing year.

Technical lock-in in AI governance is not primarily about AI systems themselves becoming harder to govern. It is about the governance infrastructure becoming harder to bridge. Every year that the EU's conformity assessment ecosystem develops along EU-specific lines without building in mutual recognition architecture, the EU notified bodies and their processes become more tightly integrated with EU-specific documentation requirements, EU-specific risk classification logic, and EU-specific audit methodology — and correspondingly more difficult to align with frameworks developed elsewhere. Every year that NIST's AI RMF generates a body of US voluntary practice without explicit regulatory mapping to other frameworks, US companies develop compliance cultures and documentation architectures that are implicitly NIST-shaped — and correspondingly resistant to audit by EU notified bodies. Every year that China's national AI standards diverge from ISO/IEC JTC 1/SC 42 outputs, the technical substrate on which any future interoperability must be built grows more heterogeneous.

This is not pessimism — it is an observation about how institutional ecosystems work. Governance infrastructure, once built, has significant inertia. The conformity assessment bodies that will dominate AI audit practice in 2030 are being established now. The documentation standards that auditors will use are being drafted now. The bilateral and multilateral agreements that will govern cross-border AI liability are, or should be, under preliminary negotiation now. Waiting for a comprehensive multilateral agreement before addressing these specific, bounded infrastructure questions is not prudent patience — it is allowing the window to close through inaction.

We have argued in this paper that the fragmentation of global AI governance is a structurally self-reinforcing system, that it creates compounding governance failures with concrete and severe costs, that the conventional response of seeking harmonization is the wrong goal, and that managed interoperability — technically demanding but institutionally achievable — is the right target. We have described what managed interoperability requires: shared incident infrastructure, mutual conformity assessment recognition, common audit trail standards, a federated liability allocation mechanism, and a standing coordination function. We have offered seven specific recommendations with named actors, specified actions, and minimum viable versions designed to constitute genuine progress within the constraints that actually bind.

The problem is not that we do not know what to do. The problem is that the institutional incentives favor delay: each jurisdiction prefers to complete its own framework before

committing to interoperability obligations, each standards body prefers to publish its own standards before aligning with others, each government prefers to negotiate bilateral agreements from a position of regulatory maturity it has not yet achieved. In the meantime, the AI systems that are making consequential decisions about people's lives in consequential contexts operate in a governance vacuum, and the cost of building the bridge grows every year.

We are not in the business of optimism. We are in the business of identifying what must be done, by whom, by when. The answer to all three questions is: much more than is currently being done, by the actors named in this paper, in the next eighteen months. That is the honest conclusion. We invite the Hollow House Institute's readers, partners, and interlocutors to treat it as such.

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